Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER–VII EXAMINATION – WINTER 2025

Subject Code:3174201 **Date:13-11-2025**

Subject Name:Deep Learning: Principles and Practices

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			Marks
Q.1	**(a)**	Define a neural network and explain its biological inspiration.	**03**
	(b)	What are the main differences between AI, Machine Learning, and Deep Learning?	**04**
	(c)	Explain the difference between Deep and Shallow Networks.	**07**
Q.2	**(a)**	How do neural networks handle non-linear relationships in data?	**03**
	(b)	What is a deep feedforward network? How does it differ from other types of neural networks?	**04**
	(c)	Explain the role of activation functions in neural networks. Give examples of commonly used activation functions.	**07**
		OR	
	(c)	Explain the concept of backpropagation in neural networks and its significance in training.	**07**
Q.3	**(a)**	What is scalar and vector?	**03**
	(b)	What is Tensor Flow? What is the main feature of Tensor Flow?	**04**
	(c)	How to Prevent Overfitting and Underfitting in Deep Learning?	**07**
		OR	
Q.3	**(a)**	Briefly explain pattern recognition using ANN.	**03**
	(b)	What is the main difference between a single-layer perceptron and a multilayer perceptron?	**04**
	(c)	What is the vanishing gradient problem? How does it affect training?	**07**
Q.4	**(a)**	Explain the properties of fuzzy sets.	**03**
	(b)	Compare between crisp/classical & fuzzy sets.	**04**
	(c)	Explain the architecture of fuzzy logic control design/controller.	**07**
		OR	
Q.4	**(a)**	Describe the Fuzzy Implication Rules.	**03**
	(b)	List the applications of fuzzy logic.	**04**
	(c)	Compare and contrast different optimization algorithms used in deep neural network training, such as Momentum Optimizer, RMSProp, and Adam.	**07**
Q.5	**(a)**	How do RBFN deep learning algorithms work?	**03**
	(b)	Discuss the use of recursive neural networks in parsing and sentiment analysis tasks with examples.	**04**

(c) Explain the working principle of a convolutional neural network and discuss the different building blocks used to extract features from input data. **07**

OR

Q.5 **(a)** What is the main idea behind generative adversarial networks, and how are they used in generative modeling? **03**

(b) What is object detection, and how is it different from image segmentation? **04**

(c) What is the working principle of LSTM networks, and how are they used in deep learning applications? **07**

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